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Decoupled Identity and Attribute Tokenization for Person Re-Identification

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Abstract

Vision-language models like CLIP have revolutionized person re-identification (ReID) by enabling cross-modal semantic alignment. However, most of the existing CLIP-based ReID methods suffer from a critical limitation: semantic entanglement, where identity and attribute features are indiscriminately compressed into a single, undifferentiated token representation. This oversight fails to account for their inherently distinct roles in characterizing individuals. To address this limitation, we propose an Identity-Attribute-Decoupled Tokenization (IADT) method, a hierarchical framework with two synergistic components: Subject-oriented tokens that model identity through a cross-modality feature inverse mapping paradigm, preserving invariant biometric features; Attribute-aware tokens that capture localized characteristics through the cross-interaction of local features and learnable prototype vectors, dynamically focusing on discriminative regions without manual supervision. The hierarchical tokenization enables disentangled yet complementary representation learning: Identity and attribute semantics are encoded into distinct embedding subspaces, while cross-token contrastive learning establishes semantic reinforcement through attention-guided feature interaction. Crucially, this process does not require part-level annotations, making it directly applicable to real-world deployment. Extensive experiments validate effectiveness of the proposed method. For example, on the Market-1501 dataset, IADT achieves 97.1% mAP (+2.5% over SOTA) and 98.2% Rank-1 accuracy. For the challenging MSMT benchmark, it attains 88.9% mAP (+1.7% improvement) with 93.1% Rank-1 accuracy, demonstrating consistent superiority. The code will be available at https://github.com/llraay/IADT.

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CCS Concepts

• **Information systems → Image search**; • **Computing methodologies → Visual content-based indexing and retrieval**.

Keywords

Person re-identification, Identity and attribute tokenization, Multimodal learning

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1 Introduction

Person Re-identification (ReID), a cornerstone of intelligent surveillance systems, addresses the critical task of retrieving individuals across non-overlapping camera networks in real-world environments. As an integral component of public safety infrastructure, ReID enables cross-camera tracking in large-scale surveillance systems deployed in transportation hubs, smart cities, and retail spaces. Despite advances in deep learning, practical deployments remain challenged by severe intra-class appearance variations induced by illumination changes, viewpoint disparities, occlusions, and pose diversity[5, 7, 15–17, 38]. These factors hinder model generalizability, highlighting the need for robust feature learning frameworks that balance discriminative ability with real-world complexity.

Convolutional Neural Networks (CNNs) [13] and Vision Transformers (ViTs)[3, 8, 20] as the main backbone networks have driven advancements in person re-identification. CNNs exploit inductive biases and hierarchical convolutions to efficiently capture fine-grained spatial patterns (e.g., clothing textures). In contrast, ViTs leverage global self-attention to adaptively focus on discriminative regions (e.g., occluded body parts) and mitigate viewpoint variations [27]. CNNs struggle with occlusions/pose shifts due to limited receptive fields and redundant spatial partitioning, while ViTs sacrifice local details via patch tokenization and require extensive data to avoid overfitting. These uni-modal trade-offs (discriminative detail vs. holistic generalization) motivate multimodal approaches.

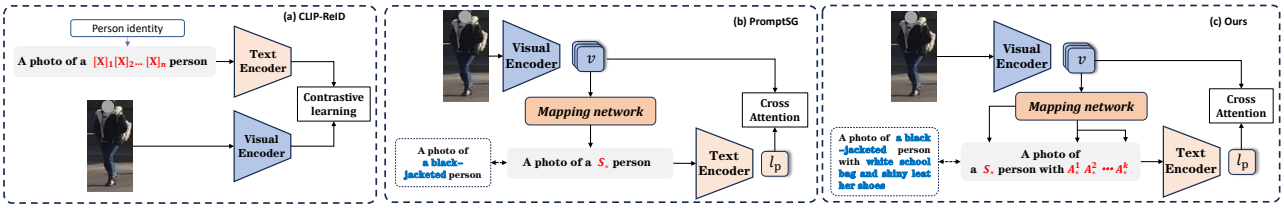

Figure 1: An illustration of CLIP-based person ReID methods. (a) CLIP-ReID [12] maps the person identities into a learnable prompt. (b) PromptSG [30] inverse the global visual features into a subject pseudo-word token. (c) Ours maps the global and local visual features into subject and attribute pseudo-word tokens, respectively.

Recent advances in Vision-Language Pre-training (VLP) models, such as CLIP [21], have unlocked novel paradigms for person re-identification. CLIP's dual-encoder architecture, pre-trained on massive image-text pairs, aligns visual and textual modalities into a unified semantic space through contrastive optimization. This enables cross-modal transfer of semantic knowledge (e.g., attribute descriptions) without requiring additional supervision. Building on this, CLIP-based ReID methods like CLIP-ReID [12] enhance identity specificity by integrating learnable text prompts—trainable vectors appended to identity-specific textual templates—to refine visual-textual alignment as shown in Figure 1(a). TSLIP-ReID [24] employs textual self-supervision to enhance identity-specific feature learning, while PromptSG [30] introduces pseudo-token learning to generate context-aware textual descriptions aligned with visual semantics as shown in Figure 1(b). Despite their advancements, these methods inherit a critical flaw from CLIP's architecture: semantic entanglement, where identity and attributes are compressed into homogeneous token representations. This conflation limits discriminative ability in complex scenarios, such as distinguishing individuals with similar attributes or resolving occlusions where localized attributes dominate identity cues.

To address these limitations of existing CLIP-based ReID methods, we propose an Identity-Attribute-Decoupled Tokenization (IADT) method that decouples identity and attribute semantics using natural language-inspired descriptors. Unlike conventional approaches that conflate these components into entangled embeddings, IADT aligns with human linguistic cognition by explicitly separating identity-centric and attribute-aware semantics as shown in Figure 1(c). Specifically, we introduce structured prompts: 'a photo of a S_* person with $A_*^1 A_*^2 \cdots A_*^k$ ', in which S_* and A_*^k denote identity and attribute tokens, respectively. This design mirrors human descriptions by assigning identity and attribute information to dedicated subspaces within the CLIP embedding space, thereby effectively avoiding semantic interference. Then, we propose an instance-specific prompt learning framework that integrates dynamic feature aggregation and context-aware filtering to achieve cross-region semantic fusion, generating context-aware fine-grained token embeddings. This framework enhances cross-modal alignment by adaptively correlating localized visual features with their textual counterparts while mitigating overfitting through attribute-conditioned regularization and noise suppression. Moreover, a lightweight hierarchical network, which uses textual embeddings derived from the prompt as the query and patch-wise embeddings from the visual encoder as the key and value, bridges the inherent gap between linguistic and visual modalities. By synergizing these components, IADT achieves deeper semantic mining: attribute tokens refine local discriminability, while identity tokens maintain global distinctiveness, overcoming the 'discriminative vs. holistic' trade-off in uni-modal ReID.

Extensive experiments on four public benchmark datasets demonstrate the superiority of our method. For example, on the Market-1501 dataset, IADT achieves 97.1% mAP (+2.5% over SOTA) and 98.2% Rank-1 accuracy. For the challenging MSMT benchmark, it attains 88.9% mAP (+1.7% improvement) with 93.1% Rank-1 accuracy, demonstrating consistent superiority.

To summarize, the contributions of our work are as follows.

- We propose an Identity-Attribute-Decoupled Tokenization method for person re-identification that decouples identity and attribute semantics using natural language-inspired descriptors.
- An instance-specific prompt optimization framework based on dynamic feature aggregation and context-aware filtering achieves cross-region semantic integration, generating context-aware fine-grained token embeddings.
- Prompt-based cross-modal learning utilizes language prompts as a guide for image feature alignment, enabling the model to localize regions corresponding to the textual descriptions.

2 Related Work

2.1 Person re-identification

Person re-identification has shown amazing promise in the field of security and protection. Thanks to the capability demonstrated by deep learning, recent work on person re-identification has opted for Convolutional Neural Network[6, 28, 31, 36] and Vision Transformer based networks, both of which are the main backbone networks in today's deep learning era. TransReID[8] is the first to bring pure Transformer introduces ReID framework, extracts features through the ViT backbone, and jointly optimizes ID loss and triplet loss, which achieves state-of-the-art results. AAFormer[44] introduces a part-level dynamic alignment mechanism within the Transformer architecture, leveraging learnable part tokens to adaptively localize semantic human/object parts without manual annotations. DC-Former[14] employs a multi-embedding space decoupling strategy, decomposing the feature space into multiple compact subspaces to enhance diversity. However, CNN-based and ViT-based ReID methods primarily rely on visual single-modality and lack the ability to integrate textual semantics, which limits the model's generalization understanding of cross-scene person attributes.

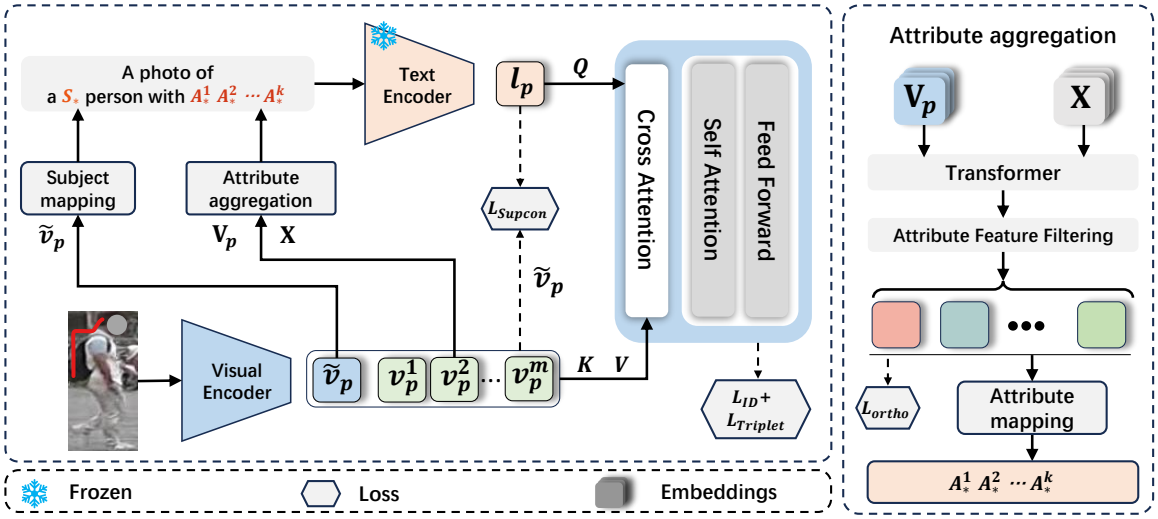

Figure 2: Overview of our proposed Identity-Attribute-Decoupled Tokenization framework. Our hierarchical architecture consists of: (1) Subject-oriented tokens generated via cross-modal feature inverse mapping to preserve identity-critical biometric patterns, and (2) Attribute-aware tokens formed through unsupervised local feature-prototype interactions that dynamically attend to discriminative regions. The dual-branch design explicitly decouples persistent identity cues from transient attribute features while maintaining their synergistic relationship through shared backbone representations.

2.2 Vision-language models for ReID

Vision-Language pre-training Models (VLMs), capable of simultaneously processing image and text information, establish semantic correlations between vision and language, demonstrating exceptional generalization abilities across many downstream tasks[1, 19]. CLIP (Contrastive Language-Image Pre-training), a landmark work in VLMs, maps images and texts into a shared embedding space through contrastive learning, achieving cross-modal semantic alignment. Inspired by recent advances in prompt learning from natural language processing (NLP)[9], CoOp[40] proposed a dynamic prompt learning framework specifically designed for pretrained models. Following this soft prompt approach, CLIP-ReID [12] pioneers the adaptation of CLIP for person ReID by classifying images into ID specific prompts. Similarly, CSDN [32] introduces bimodal learnable natural language descriptions to acquire the semantic information corresponding to visual representations of visible and infrared images. CCLNet [2] proposes a prompt learning paradigm for unsupervised visible-infrared person re-identification by establishing a learnable cluster-aware prompt for person images and obtain textual descriptions allowing for subsequent unsupervised training. Differing from CLIP-ReID, Prompt-driven Semantic Guidance (PromptSG) [30] focuses on vision-language alignment, whose goal is to exploit rich semantic information from language to explicitly control the weights assigned to each patch or region, and improve the two-stage framework by directly inverting images into the language latent space. However, its reliance on simple reverse network processing falls short of deeply capturing fine-grained human features. In contrast, our model employs a hierarchical design that decouples fine-grained information, reduces interference from heterogeneous semantic contexts, and highlights attribute-specific regions through targeted attention mechanisms.

3 Methodology

In this section, we provide a comprehensive overview of our proposed framework, as illustrated in Figure 2. Unlike existing methods [30] that project images into coarse global pseudo-word tokens, our approach introduces an Identity-Attribute-Decoupled Tokenization by decomposing visual content into two distinct types: one subject-oriented pseudo-word token and multiple attribute-aware pseudo-word tokens. Specifically, we introduce Instance-specific Prompt Learning framework that focuses on learning discriminative attribute information embedded in spatially localized regions of the image. By aggregating these region-specific features, our framework generates attribute-aware tokens that capture nuanced visual details. Moreover, inspired by [30, 32], we introduce a multi-modal patch-to-prompt interaction through a language-guided cross-attention module that aggregates the attention map to emphasize regions with high semantic relevance, allowing the model to better align visual and textual information. This design addresses the limitations of prior work by explicitly disentangling holistic semantic concepts from localized attribute-specific details.

3.1 Instance-specific Prompt Learning

In conventional classification tasks, CLIP achieves strong performance by mapping discrete class labels to semantically rich text prompts, a process contingent on manually engineered prompt templates. In contrast, person ReID operates under an open-set recognition framework, where identities are inherently unbounded and lack predefined class labels. This fundamental discrepancy poses a significant barrier: the absence of explicit semantic categories renders conventional CLIP-style prompt engineering inapplicable, as no textual descriptors exist to align visual features with identity-specific semantics. To address this, we propose using visual features

as guidance to generate adaptive subject/attribute tokens. Based on this, we introduce structured prompts: 'a photo of a S_* person with $A_*^1 A_*^2 \cdots A_*^k$', in which S_* and A_*^k denote learnable identity and attribute tokens, derived from visual features. Specifically, this structure mimics human-centric descriptions while enforcing a subspace disentanglement strategy: identity and attribute tokens are optimized to occupy orthogonal subspaces within CLIP embedding space, thereby eliminating semantic interference between categorical (identity) and descriptive (attribute) features.

3.1.1 Subject-oriented Pseudo-word Token. Following the prior works, we design a subject mapping network f_S to project the global visual embedding $\tilde{v}_p \in \mathbb{R}^d$ from CLIP's visual encoder into a pseudo-language token in the textual embedding space, i.e., $S_* = f_S(\tilde{v}_p)$. This token acts as a context-aware semantic anchor, encoding subject-level semantics of the input image while ensuring tight alignment between the visual embedding $\tilde{v}_p$ and its textual counterpart S_* in CLIP's shared latent space. By enforcing cross-modal consistency, the proposed approach yields a discriminative representation that preserves the unique identity characteristics of the input, thereby effectively bridging the modality gap between vision and language.

3.1.2 Attribute-aware Pseudo-word Token. Prior research demonstrates that global features struggle to reliably identify individuals under occlusions, viewpoint variations, and other real-world complexities in person re-identification, as they lack fine-grained discriminative cues. While our initial pseudo-language identity token S_* anchors subject-level semantics in CLIP's shared latent space, treating S_* as a monolithic 'natural language word' risks conflating heterogeneous semantics. For instance, human descriptions prioritize specificity (e.g., 'an elegant person wearing a suit and bowler hat') over vague abstractions (e.g., 'an elegant person'). To address this, we propose attribute-aware pseudo-word tokens A_*^k, which encode fine-grained local attributes by adaptively aggregating semantically salient image patches. Unlike generic open-domain attribute models, our method prioritizes person-specific attributes cues through a dynamic patch fusion mechanism. This mechanism employs a domain-specific latent attribute to guide the aggregation of localized features while suppressing irrelevant background noise via a local-global relevance filtering strategy.

Attribute extraction: Let $\mathbf{V}_p = \left\{v_p^i\right\}_{i=1}^m \in \mathbb{R}^{d\times m}$ denote the patch-wise visual features extracted from the penultimate layer of the CLIP visual encoder, d denotes the dimension of the features, and m denotes the number of image patches. Consequently, to capture latent attribute information, we initialize a set of learnable query embeddings $\mathbf{X} \in \mathbb{R}^{d\times n}$, which correspond to the n latent local attribute features of the images. To adaptively learn discriminative local attribute features, we employ a standard Transformer network, which leverages self-attention mechanisms to model comprehensive interactions between input elements. The network processes the combined input of learnable query embeddings $\mathbf{X} \in \mathbb{R}^{d\times n}$ (representing latent attribute prototypes) and visual patch features $\mathbf{V}_p = \left\{v_p^i\right\}_{i=1}^m \in \mathbb{R}^{d\times m}$. Through multi-head self-attention layers, the Transformer effectively captures spatial-contextual relationships and outputs refined local attribute representations. To enable effective local-global relevance-based filtering, we introduce a dimension alignment module consisting of a fully-connected layer that projects the Transformer's output attributes to match the dimension of the global image feature $\tilde{v}_p$. Formally, this process can be expressed as follows,

$$\mathbf{X}' = FC\left(\mathcal{F}_{\text{Transformer}}\left([\mathbf{X}|\mathbf{V}_p]\right)\right) \tag{1}$$

where $[\cdot|\cdot]$ represents cascade operation, FC denotes the fully connected network, and $\mathbf{X}' = \left\{x_i'\right\}_{i=1}^n \in \mathbb{R}^{d\times n}$ denotes the local attribute features of the image.

Attribute filtering: While our adaptive local feature aggregation effectively captures discriminative attributes, we identify two inherent challenges: (1) Background noise contamination from repetitive or irrelevant regions (e.g., recurring background patterns), and (2) Semantic overlap between local attribute features and global features, where attribute descriptors often elaborate on aspects already partially encoded in the global representation.

To mitigate the noise amplification and feature redundancy inherent in adaptive local feature aggregation, we propose a principled attribute selection mechanism grounded in local-global semantic relevance. For each projected local attribute feature x_i' and global image feature $\tilde{v}_p$, we calculate normalized relevance scores $s_i = \tilde{v}_p^\top \cdot x_i' / (\|x_i'\| \cdot \|\tilde{v}_p\|)$. Then ranks attributes by descending relevance and finally retains only the top-k features $\mathbf{W} = \left[x_j'\right], s_j \in \text{top-}k\,(S)$ where s_j indicates the semantic relevance between the global image feature $\tilde{v}$ and the j-th learned local attribute feature x_i'. Top-$k(S)$ denotes the top k entry values of S, where $S = [s_1, s_2, \cdots, s_n]$. $\mathbf{W} \in \mathbb{R}^{d\times k}$ represents the selected valid local attribute features.

Attribute refining: To ensure distinct semantic representations among the local attribute features, we introduce an orthogonal constraint on the selected attribute feature matrix $\mathbf{W}$, imposing maximal discrimination between learned attributes. Formally, the orthogonal regularization loss is defined as:

$$\mathcal{L}_{\text{ortho}} = \left\|\mathbf{W}\mathbf{W}^\top - \mathbf{I}\right\|_F^2 \tag{2}$$

where $\mathbf{I}$ denotes the identity matrix. This formulation yields two key benefits: (1) it promotes diversity among attribute embeddings by minimizing inter-feature correlations, and (2) it implicitly suppresses interference from irrelevant features during learning. The constraint acts as a regularizer that maintains the geometric independence of attribute directions in the latent space while preserving the model's ability to focus on salient visual characteristics.

Given the inherent differences between the global image feature and local attribute features, we employ another MLP-based attribute mapping network, denoted as $f_A(\cdot)$, to project the local attribute features into the real-word token embedding space. Mathematically, we have

$$\left[A_*^1, A_*^2, \cdots A_*^k\right] = f_A(\mathbf{W}) \tag{3}$$

By combining the subject and attribute-oriented tokens with a generic textual prompt, we construct an adaptive language prompt tailored to each image: 'a photo of a S_* person with $A_*^1 A_*^2 \cdots A_*^k$'. This prompt is tokenized into t_p and processed by CLIP's text encoder, producing the text embedding l_p.

To guarantee that the learned text embedding l_p effectively represents the image semantics and maintains identity consistency,

we employ symmetric supervised contrastive loss to encourage the text embedding to capture unique visual details.

$$\mathcal{L}_{i2t}^{\text{Sup}} = \frac{1}{N}\sum_{n=1}^{N}\sum_{p^+\in P(n)} \log \frac{\exp(\text{sim}(v_n, l_{p^+})/\tau)}{\sum_{i=1}^{N}\exp(\text{sim}(v_n, l_i)/\tau)},$$
$$\mathcal{L}_{t2i}^{\text{Sup}} = \frac{1}{N}\sum_{n=1}^{N}\sum_{p^+\in P(n)} \log \frac{\exp(\text{sim}(l_n, v_{p^+})/\tau)}{\sum_{i=1}^{N}\exp(\text{sim}(l_n, v_i)/\tau)}, \tag{4}$$

$$\mathcal{L}_{SupCon} = \mathcal{L}_{i2t}^{\text{Sup}} + \mathcal{L}_{t2i}^{\text{Sup}}, \tag{5}$$

where $P(n)$ denotes the positive samples associated with v_n and l_n.

3.2 Prompt-based Cross-Modal Learning

To enhance fine-grained semantic alignment, we further introduce the Prompt-based Cross-attention strategy. The core idea of our approach is to take advantage of language prompts as a guide for image feature alignment, enabling the model to localize regions corresponding to the textual descriptions. Specifically, image regions associated with the semantic concept of 'person' and identity-defining attributes are prioritized, as these elements are critical for discriminative cross-modal matching.

In order to realize the multimodal intersection of text and image, and guide the model to focus on fine-grained information about the subject and the attribute, we introduce a multimodal attention module, which uses the fused text embedding l_p as the query Q, and the image-based visual embedding $\left\{\tilde{\boldsymbol{v}}_p, \boldsymbol{v}_p^1, \ldots, \boldsymbol{v}_p^m\right\}$ as keys K and values V. This process can be represented as follows.

$$\text{A}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{6}$$

This interaction aggregates the attention map to highlight regions with high semantic responses. Inspired by multimodal fusion methods [4], we introduce two transformer blocks following the cross-attention layer to refine the final representations. The framework is optimized using the standard ReID loss, which includes the triplet loss and the identity classification loss.

$$\mathcal{L}_{\text{ID}} = \frac{1}{K}\sum_{j=1}^{K} y_j \log p_j \tag{7}$$

$$\mathcal{L}_{\text{Triplet}} = \max\left(d_p - d_n + m, 0\right) \tag{8}$$

where p_j is the prediction probability for the j-th class, and m denotes the margin.

In summary, the total loss of our framework is as follows.

$$\mathcal{L} = \mathcal{L}_{\text{Triplet}} + \mathcal{L}_{\text{ID}} + \mathcal{L}_{\text{ortho}} + \lambda\mathcal{L}_{\text{Supcon}} \tag{9}$$

4 Experimental Results

4.1 Experimental Settings

4.1.1 Datasets and Evaluation Protocols. In this section, we evaluate the performance of our IADT on two ReID tasks: the general ReID and occluded ReID on four widely adopted ReID datasets: Market-1501[37], MSMT17[29], DukeMTMC-reID[39] and Occluded-Duke[18]. Partial information about these datasets is summarized in Table 1. Rank-1 (R1) accuracy from the Cumulative Match Characteristic (CMC) curve and mean Average Precision (mAP) are employed as evaluation metrics to assess the model's performance .

Table 1: The statistics of datasets in our experiments

Dataset	#ID	Images	Cams
Market-1501	1,501	32,668	6
MSMT17	4,101	126,441	15
DukeMTMC	1,404	36,411	8
Occluded-Duke	1,404	35,489	8

4.1.2 Implementation Details. For fair comparison with previous works, we employ CLIP's pretrained ResNet-50 and ViT-B/16 models as visual encoders. The text encoder maintains a fixed Transformer architecture (CLIP text Transformer). All person images are resized to 256×128 resolution and augmented by horizontal flipping and random cropping. Training batch size is set to 64, containing 16 distinct identities with 4 images per identity. We utilize the Adam optimizer, setting the learning rate at 5e-6 for visual encoders and 5e-5 for randomly initialized modules. A 10-epoch warm-up strategy linearly increases the learning rate from 5e-7 to 5e-6. For CNN-based models, training spans 120 epochs with learning rate decayed by 0.1 at epochs 40 and 70, while ViT-based models train for 60 epochs with 0.1 decay every 20 epochs. We set the number of selected attributes to 16 and 64 for learnable queries in the attribute-aware pseudo-word tokens. As mentioned earlier, we employ two three-layer MLPs (Multi-layer Perceptrons) f_S and f_A with 512-dimensional hidden layers to map the subject and attribute tokens, respectively. Additionally, a three-layer single-head Transformer is utilized to extract image attribute features. The framework incorporates a Transformer-based dynamic attribute feature aggregation module. In Eq. (9), hyper-parameter λ is set to 0.1. All experiments are implemented using PyTorch. Comparative studies with state-of-the-art (SOTA) methods on DukeMTMC and Occluded-Duke datasets are conducted on a single NVIDIA 2080Ti GPU, while other experiments use a single NVIDIA 3090 GPU.

4.2 Comparison to the SOTA Methods

We evaluate the proposed IADT against state-of-the-art methods, systematically categorizing existing approaches into CNN-based, ViT-based, and CLIP-based paradigms. As detailed in Table 2, IADT achieves consistent superiority across all four benchmarks, with margins of +2.5–8.5% mAP over the best-performing counterparts. Notably, our method demonstrates robust cross-architecture generalization, outperforming both ViT and CNN backbones.

To verify the generalizability and performance advantages of the presented method, we perform a side-by-side comparison with the current optimal method on four mainstream datasets, namely Market1501, MSMT17, DukeMTMC-reID and Occluded-Duke. The experiments show that the proposed method exhibits significant advantages in cross-scene, multi-scale and complex interference conditions: as shown in Table 2 on the Market1501 dataset, the proposed method outperforms the existing optimal model (+1.2% Rank-1, +2.5% mAP) with 98.2% Rank-1 and 97.1% mAP, which

Table 2: Comparison with the state-of-the-art models on Market-1501, MSMT17, DukeMTMC, and Occluded-Duke datasets. The superscript star* indicates that the image is resized to a resolution exceeding 256x128.

Backbone	Methods	References	Market-1501		MSMT17		DukeMTMC		Occluded-Duke	
			mAP	R1	mAP	R1	mAP	R1	mAP	R1
ResNet50	CNN-based method									
	OSNeT[41]	ICCV'19	84.9	94.8	52.9	78.7	73.5	88.6	-	-
	HOReID[25]	CVPR'20	-	-	84.9	94.2	75.6	86.9	43.8	55.1
	ISP[43]	ECCV'20	84.9	94.2	-	-	75.6	86.9	52.3	62.8
	RGA-SC[35]	CVPR'20	88.4	96.1	57.5	80.3	-	-	-	-
	CDNet[10]	CVPR'21	86.0	95.1	54.7	78.9	76.8	88.6	-	-
	CAL[22]	ICCV'21	87.0	94.5	56.2	79.5	76.4	87.6	-	-
	ALDER*[34]	TIP'21	88.9	95.6	59.1	82.5	78.9	89.9	-	-
	LTReID*[26]	TMM'22	86.9	94.7	58.6	81.0	80.4	**90.5**	-	-
	CLIP-based method									
	Baseline		88.1	94.7	60.7	82.1	79.3	88.6	47.4	54.2
	CLIP-ReID[12]	AAAI'23	89.8	95.7	63.0	84.4	80.7	90.0	53.5	61.0
	PromptSG[30]	CVPR'24	91.8	96.6	68.5	86.0	80.4	90.2	54.0	59.1
	IADT	Ours	**94.4**	**96.8**	**83.1**	**87.5**	**86.0**	**90.5**	**62.2**	**66.2**
ViT-B/16	ViT-based method									
	TransReID[8]	ICCV'21	88.9	95.2	67.4	85.3	82.0	90.7	59.2	66.4
	DCAL[42]	CVPR'22	87.5	94.7	64.0	83.1	80.1	89.0	-	-
	AAformer[44]	TNNLS'23	88.0	95.4	65.6	84.4	80.9	90.1	58.2	67.0
	PHA[33]	CVPR'23	90.2	96.1	68.9	86.1	-	-	-	-
	CLIP-based method									
	Baseline		86.4	93.3	66.1	84.4	80.0	88.8	53.5	60.8
	CLIP-ReID[12]	AAAI'23	89.6	95.5	73.4	88.7	82.5	90.0	59.5	67.1
	PromptSG[30]	CVPR'24	94.6	97.0	87.2	92.6	81.6	91.0	66.4	68.4
	IADT	Ours	**97.1**	**98.2**	**88.9**	**93.1**	**91.0**	**94.2**	**70.9**	**74.6**

verifies the effectiveness of the cross-modal feature fusion strategy. On the large-scale real scene dataset MSMT17, the proposed method improves 0.6% and 1.7% on Rank-1 (93.1%) and mAP (88.9%), respectively, demonstrating its strong robustness to illumination variations and viewpoint differences. In addition, our framework also demonstrates superior performance when adopting CNN as the backbone. On the MSMT17 dataset, an improvement of about 15% for mAP is achieved, which is almost on par with existing ViT-based methods. The above results consistently show that our subject-attribute decoupled and alignment strategy achieves a balanced performance improvement across datasets without relying on fine-grained manual annotation or complex pre-training, and provides a better solution for the task of person re-identification.

4.3 Ablation Studies

To validate the necessity of each component in IADT, we perform ablation studies on the Market-1501 and MSMT17 datasets under identical ViT-B/16 backbone configurations. We systematically evaluate two critical aspects: (1) Instance-specific prompt, (2) Effect of attribute-aware tokens.

4.3.1 Ablation studies on the instance-specific prompt. The ablation study in Table 3 evaluates the contribution of key components—subject tokens (S_*) and attribute prompts (A_*^k)—on Market-1501 and MSMT17. When neither component is used, i.e., 'a photo of a person' (row a), baseline performance reaches 92.0% mAP / 96.3% R-1 on Market-1501 and 85.3% mAP / 91.6% R-1 on MSMT17. Introducing the subject token (S_*) alone, i.e., 'a photo of a S_* person' (row b) yields notable improvements: +2.6% mAP and +0.7% R-1 on Market-1501, and +1.9% mAP / +1.0% R-1 on MSMT17, demonstrating its role in enhancing feature discrimination. Further incorporating attribute prompts (A_*^k), i.e., our structured prompt 'a photo of a S_* person with $A_*^1 A_*^2 \cdots A_*^k$' (row c) elevates performance to 97.1% mAP / 98.2% R-1 and 88.9% mAP / 93.1% R-1, highlighting the complementary effect of attribute-level cues. This validates the necessity of both components for robust person re-identification.

Table 3: Ablation study on training with structured prompts of varying compositions on Market-1501 and MSMT17.

	Components		Market-1501		MSMT17	
	S_*	A_*^k	mAP	R-1	mAP	R-1
a)			92.0	96.3	85.3	91.6
b)	✓		94.6	97.0	87.2	92.6
c)	✓	✓	97.1	98.2	88.9	93.1

4.3.2 Analysis of attribute-aware pseudo-word tokens. To further analyze the effectiveness of our attribute-aware pseudo-word tokens, in this section, we conduct two experiments to validate the

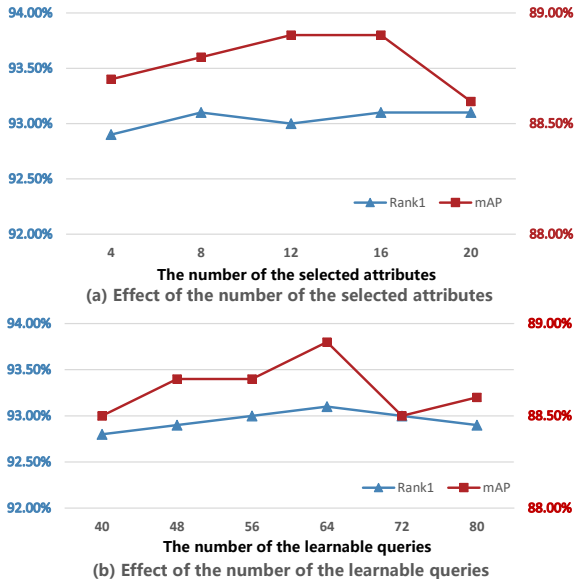

Figure 3: Analysis of influential factors of attribute-aware pseudo-word tokens. (a) Effect of the number of the selected attributes, (b) Effect of the number of the learnable queries.

effect caused by the various numbers of selected attributes and learnable queries on the challenging MSMT17 dataset.

Effect of the number of the selected attributes. Figure 3 (a) investigates the impact of varying the number of selected attributes. As the attribute count increases to 16, mAP exhibits progressive improvements. This underscores the critical role of attribute diversity in addressing MSMT17's inherent complexities, as richer attribute sets capture finer visual cues to disambiguate identities under variable real-world conditions. However, the diminishing marginal gains beyond 16–20 attributes suggest a saturation point, where additional features contribute minimally to performance, and unavoidable interference caused by background noise. Throughout this process, the Rank-1 metric demonstrates strong robustness. This highlights a trade-off: sufficient attributes are necessary for fine-grained semantic alignment, but excessive attributes introduce background noise interference, leading to performance degradation.

Effect of the number of the learnable queries. Figure 3 (b) analyzes the effect of learnable query counts on MSMT17. Increasing queries (40-64) boosts Rank-1 accuracy, as more queries capture fine-grained identity cues in cluttered scenes. The mAP peaks at 64 queries (88.9%) and slightly declines, suggesting excessive queries introduce noise, reducing retrieval consistency. This divergence highlights a trade-off: higher queries improve top-match precision but risk overfitting to transient patterns. In the experiments, we set the number of learnable queries to 64.

4.4 Visualization of IADT

4.4.1 Attention map visualization. To enable more intuitive analysis of our model's effectiveness, we visualize the model's attention. We adopt the general Grad-CAM[23] method to facilitate universal comparisons with other approaches. Figure 4 visualizes Grad-CAM attention maps to compare our full IADT model with baselines

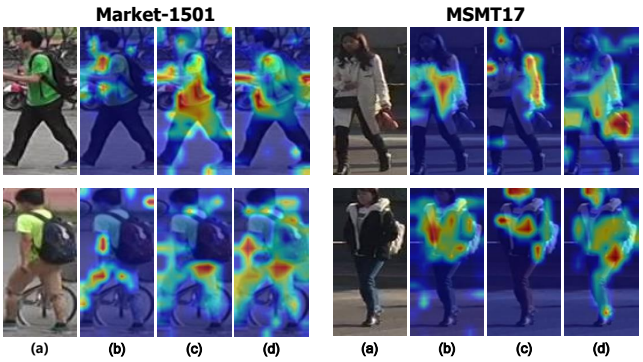

Figure 4: Grad-CAM visualization of attention maps. (a) Original images, (b) PromptSG, (c) IADT without A_*^k training, (d) IADT. We see our method is effective in simultaneously focusing on the semantic clues and exploring more discriminative parts.

Figure 5: Comparison of top-10 retrieved results for PromptSG (the first row) and our IADT (the second row) for each query. The incorrectly identified samples are highlighted in red, while the green highlights the correct samples.

and ablated variants on the MSMT17 dataset. The original images (a) highlight challenging scenarios, such as occluded body parts or cluttered backgrounds. PromptSG (b) exhibits diffuse attention, often focusing on non-discriminative regions (e.g., background artifacts), which limits its ability to capture identity-critical features. The ablated IADT variant without A_*^k training (c) shows partial improvement, localizing coarse semantic regions (e.g., torso or head) but missing fine details (e.g., bags or footwear). In contrast, our full IADT model (d) generates precise, high-resolution attention maps that prioritize both semantic coherence (e.g., clothing patterns) and discriminative attributes (e.g., accessories or unique textures), even under occlusion. This demonstrates the necessity of A_*^k training,

which enables adaptive multi-scale attribute learning, guiding the model to suppress noise and amplify subtle identity cues. The visual results align with quantitative improvements in Tables 2–3, where IADT achieves superior Rank-1/mAP scores, underscoring that robust attention to discriminative regions directly translates to enhanced re-identification accuracy in real-world conditions.

4.4.2 Retrieval result visualization. This study conducts cross-camera person retrieval experiments on two benchmark datasets, Market1501 and MSMT17, specifically selecting challenging query images with substantial interference factors to validate the model's fine-grained feature extraction capability in extreme scenarios. As illustrated in Figure 5, the first row demonstrates retrieval results from the baseline method PromptSG, while the second row presents outcomes of our proposed IADT approach.

In the first test case, despite severe motion blur causing significant degradation of pedestrian contours and textural details in the query image, our method successfully identifies the correct target. The second more challenging scenario involves a query image with indistinguishable main body information, where our model effectively focuses on discriminative attribute features - particularly a handbag - to achieve accurate recognition. It can be observed that PromptSG also notices similar but less precise information, resulting in the misrecognition of a highly similar individual. These observations substantiate that our method demonstrates enhanced attentional sensitivity to fine-grained discriminative features.

4.5 Pseudo-to-Real Description Retrieval

To validate the semantic alignment of our learned subject-oriented S_* and attribute-aware $[A_*^1, ..., A_*k]$ pseudo-word tokens with natural language, we propose a cross-modal retrieval framework. Leveraging BLIP [11]), we construct a 10k-entry real-language attribute corpus from pre-training data after filtering out the generic phrases (e.g., "blurry photo") on the Market1501 dataset. Each image is encoded as a pseudo-text prompt ("a photo of a S_* person with $[A_*^1, ..., A_*^k]$") within CLIP's embedding space, and the most related description from the top-20 results is retrieved via cosine similarity. Considering the person re-identification task, all the subject descriptions are 'person', therefore, we focus more on the retrieval performance regarding the attribute descriptions. As illustrated in Figure 6, S_* tokens reliably map to person-centric contexts (e.g., "walking down the street"), while $[A_*^1, ..., A_*^k]$ tokens capture discriminative attributes (e.g., "striped shirt", "holding a bag"). This demonstrates that pseudo-tokens not only encode human-interpretable semantics but also bridge the latent space between visual representations and textual descriptions, a critical advancement for explainable cross-modal ReID systems.

4.6 Hyperparameter Analyses

As shown in the Figure 7, we analyze the impact of the hyperparameter λ in the Eq. 9 on mAP and Rank-1 metrics. It can be seen that IADT achieves the best performance at $\lambda = 0.1$. Increasing λ led to a general decline in both mAP and Rank-1. The performance decline at higher λ values potentially stems from over-alignment: CLIP's contrastive loss, when excessively weighted, rigidly forces image-text alignment in ReID's high inter-class similarity setting, suppressing subtle visual distinctions. Limited textual diversity

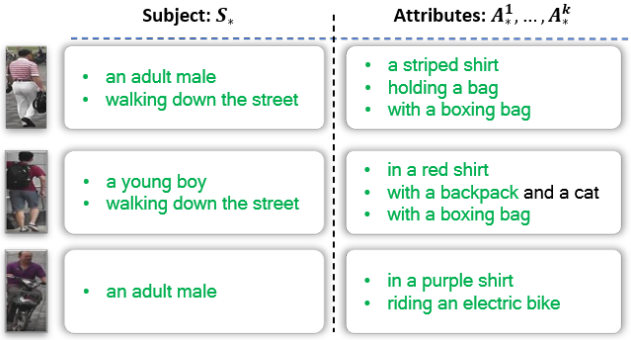

Figure 6: Pseudo-to-real description retrieved results. We highlight the related real-word descriptions in green.

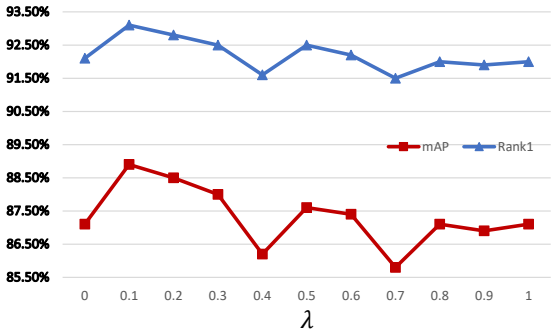

Figure 7: The influence of different weights of λ on the mAP and Rank-1 metrics on MSMT17.

(e.g., generic ID labels) fails to guide nuanced feature separation, while dominant contrastive loss overshadows discriminative objectives, collapsing features of visually similar classes. CLIP's general-purpose design, optimized for broad cross-modal tasks, struggles to preserve fine-grained ReID-specific boundaries under aggressive alignment. Optimal performance likely requires balancing λ to maintain alignment without sacrificing granularity.

5 Conclusion

This paper proposes an effective framework called Identity-Attribute-Decoupled Tokenization (IADT) for person re-identification tasks, built upon the CLIP model. Unlike existing solutions, IADT integrates subject-oriented and attribute-aware tokens to address the semantic entanglement limitation of current methods through identity-attribute disentangled tokens. To capture fine-grained information and suppress background noise, we employ an instance-specific prompt learning strategy combined with a local-global relevance-based filtering mechanism. Furthermore, by designing a structured text prompt template—"a photo of a S_* person with $A_*^1 A_*^2 \cdots A_*^k$"—we guide the model to focus on discriminative features and achieve state-of-the-art recognition performance on four challenging ReID datasets. This framework demonstrates the effectiveness of identity-attribute disentanglement and fine-grained feature learning for robust cross-modal person re-identification.

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